

PES UNIVERSITY, BENGALURU

Department of Computer Science and Engineering

SOFTWARE ENGINEERING – LEVEL 3 MINIPROJECT

SYNOPSIS / PROJECT PROPOSAL

Electronics Store Inventory Management System

Group	Team 9
Project Type	Software Engineering Level 3 Miniproject
Proposed Case Study	TechZone Electronics (Fictional Electronics Retail Store)
Development Methodology	Agile
Proposed Technology	Django REST Framework + React + PostgreSQL
Version	1.0
Date	01 September 2026

1. Project Information

The proposed system is a web-based inventory management application for a small-to-medium electronics retail store, focused on product, stock, purchase and sales management.

Sl. No.	SRN	Student Name	Primary Functional Ownership
1	PES1UG24CS068	Annamalai N	User Authentication & Role Management
2	PES1UG24CS077	Apurv Kumar Singh	Product & Inventory Management
3	PES1UG24CS091	Atharv Mittal (Team Lead)	Supplier & Purchase Management
4	PES1UG24CS099	Avaneesh H D	Sales & Reporting Management

Case Study Context

TechZone Electronics is a fictional electronics retail shop used as the case study. It sells common electronics and accessories and uses the system for internal store operations.

2. Problem Statement

Small and medium-sized electronics stores may rely on manual records or disconnected spreadsheets, making it difficult to maintain accurate stock, track purchases and sales, identify low-stock items and prepare reports.

For TechZone Electronics, these limitations can cause stock discrepancies, delayed replenishment, overstocking and unnecessary administrative effort. The proposed system centralizes products, inventory, suppliers, purchases and sales, with role-based access, low-stock identification and reporting.

The proposed Electronics Store Inventory Management System will address these issues by providing a centralized platform for managing products, inventory, suppliers, purchases and sales. The system will update stock based on valid purchase and sales transactions, identify products that reach their configured minimum stock level, provide role-based access, and generate useful inventory and transaction reports.

3. Proposed Project Description

The system will be a web application for authorized store employees, maintaining centralized product and transaction data with role-specific functions.

Major Operating Flow

- Product setup: Staff manage products, categories, SKUs, prices and minimum stock levels.
- Purchasing: Staff manage suppliers, purchase orders and received purchases.
- Inventory: Received purchases increase stock; sales decrease stock.
- Monitoring: The system identifies products at or below their minimum stock level.
- Reporting: Authorized users view inventory, purchase, sales and stock-movement reports.

Key Differentiation of the Proposed System

The system integrates purchases and sales with inventory updates, while role-based access and low-stock monitoring connect daily transactions with management visibility.

4. Target Users / Stakeholders

User / Stakeholder	Role in the System	Main Activities
Store Administrator / Manager	Primary management user	Manage users and roles, monitor inventory, review reports, manage product and supplier information
Inventory Staff	Operational inventory user	Manage products, receive purchases, update stock, monitor low-stock items
Sales Staff	Sales transaction user	View product availability, record sales/stock-out transactions, view relevant transaction information
Suppliers	External stakeholder	Provide products to the store through purchase transactions; supplier details are maintained in the system

5. Objectives

1. To maintain centralized and accurate records of electronics products, categories and stock quantities.
2. To track inventory changes resulting from purchases, sales and authorized stock adjustments.
3. To manage supplier information and purchase transactions in a structured manner.
4. To identify products whose available stock reaches or falls below a configured minimum threshold.
5. To provide authorized users with inventory, purchase, sales and stock-movement reports.
6. To reduce manual effort and improve the consistency, traceability and efficiency of routine inventory operations.

6. Scope

6.1 In-Scope

- User authentication and role-based access control.
- Product and category creation, modification, deactivation and search.
- Inventory quantity and minimum-stock management.
- Stock-in and stock-out transactions.
- Supplier information management.
- Purchase order creation and receipt recording.
- Sales transaction recording.
- Low-stock identification and alerts.
- Inventory, purchase, sales and stock-movement reports.

6.2 Out-of-Scope

- Customer-facing online shopping or e-commerce functionality.
- Online payment gateway integration.
- Home delivery, logistics or shipment tracking.
- Full accounting, taxation or ERP functionality.
- AI-based demand forecasting or automated procurement decisions.
- Enterprise-scale multi-store management.

7. Functional Features

ID	Functional Feature	Description
F-01	User Authentication & Role Management	Authenticate users and enforce role-based access to system functions.
F-02	Product & Category Management	Create, update, deactivate, search and view product/category information.
F-03	Inventory Management	Maintain current stock, minimum stock thresholds, stock adjustments and stock history.
F-04	Stock-In Management	Increase inventory when valid purchased goods are received.
F-05	Supplier Management	Maintain supplier profiles and supplier status.
F-06	Purchase Order Management	Create and manage purchase orders and record received purchases.
F-07	Sales / Stock-Out Management	Record sales and reduce inventory based on valid transactions.
F-08	Low-Stock Alert Management	Identify products at or below their configured minimum stock level.
F-09	Inventory & Transaction Reporting	Generate inventory, purchase, sales and stock-movement reports.
F-10	Search & Filtering	Allow users to locate products and transactions using relevant search/filter criteria.

8. Team-wise Functional Ownership

Each member owns one complete business functionality while all members participate in requirements, design, implementation, testing, reviews and documentation.

Member	Primary Ownership	Responsibilities / Deliverables
Annamalai N	User Authentication & Role Management	Login/logout, user accounts, roles, access permissions, authentication-related requirements and tests.
Apurv Kumar Singh	Product & Inventory Management	Products, categories, SKU data, stock quantities, minimum stock thresholds, stock adjustments and inventory views.
Atharv Mittal	Supplier & Purchase Management	Supplier profiles, purchase orders, purchase items, receiving purchases and purchase history. Integration with inventory update.
Avaneesh H D	Sales & Reporting Management	Sales/stock-out transactions, sales history, dashboards, inventory/transaction reports and reporting views.

Ownership Boundaries and Integration

Apurv manages current product stock, while Atharv manages purchases that bring stock into the store. Received purchases increase inventory, and Avaneesh's sales transactions reduce it.

9. Plan of Work

The project will use Agile development through GitHub and Jira, with each sprint producing a demonstrable and testable increment.

Sprint	Focus	Planned Output
Sprint 1	Requirements & Foundation	Finalize scope and requirements; GitHub/Jira setup; initial architecture; database foundation; authentication.
Sprint 2	Products & Inventory	Product/category management, inventory views, stock rules, stock adjustments and low-stock logic.
Sprint 3	Suppliers & Purchases	Supplier management, purchase orders, purchase receipt workflow and inventory integration.
Sprint 4	Sales & Reporting	Sales/stock-out transactions, dashboards, reports and search/filtering.
Sprint 5	Integration & Quality	Module integration, unit/integration tests, security validation, code review, CI/CD and defect fixing.
Sprint 6	System Validation & Submission	System validation, requirements traceability, documentation, maintenance plan and final demo preparation.

Project Management and Collaboration

- GitHub: version control, branches, pull requests and code reviews.
- Jira: backlog, user stories, sprint planning and issue tracking.
- Each member will contribute through feature branches, commits, pull requests and reviews.
- Testing and validation will be linked to functional requirements.
- Requirements will be traced through design, implementation and testing.

10. Proposed Technology / Tools

Area	Proposed Technology / Tool	Purpose
Frontend	React + JavaScript	Responsive web interface and role-specific application screens.
Backend	Python + Django REST Framework	REST APIs, business logic, authentication and server-side processing.
Database	PostgreSQL	Persistent storage for users, products, inventory, suppliers and transactions.
Version Control	Git + GitHub	Source control, branching, pull requests and code review.
Agile Management	Jira	User stories, backlog, sprint planning and issue tracking.
Testing	pytest + API testing tools	Unit and API-level testing and validation.
CI/CD	Jenkins	Automated build and test pipeline; deployment workflow as applicable.
Containerization	Docker	Consistent development and deployment environments.
Static Analysis	SonarQube or similar	Code-quality and static-analysis checks.

11. Preliminary Business Rules

- Only authenticated users may access protected system functions.
- Users may perform only operations permitted by their role.
- Each active product shall have a unique SKU.
- A valid received purchase shall increase the stock quantity of the corresponding products.
- A valid sale shall decrease the stock quantity of the corresponding products.
- The system shall not allow a normal sales transaction to reduce available stock below zero.

- A product shall be considered low stock when its available quantity is less than or equal to its configured minimum stock threshold.
- Deactivated products shall not be available for new transactions.
- Purchase and sales transactions shall retain transaction identifiers and timestamps for traceability.

12. Expected Outcome

The team expects to deliver a working web prototype covering products, suppliers, purchases, sales, inventory, low-stock identification and reports, supported by project records, tests and validation evidence.

13. Conclusion

The system provides a manageable case study for applying the software development life cycle, with four functional ownership areas and integrated inventory processes. Its scope is limited to internal store operations to support focused design, implementation, testing and traceability.